How a Job Portal Benefited from Microservices Architecture

Microservices architecture has emerged as a preferred pattern for building scalable and maintainable software applications. Here's how a monolithic job portal application was re-engineered into a microservices-based system. The migration process, key design decisions, the technology stack used, and measurable improvements in performance and flexibility are all laid out for you.

As an application scales, it must migrate from a monolithic architecture to a microservices-based one. Here, we document the migration of a job portal from a monolithic architecture to a microservices-based design. Initially, all core features—job listings, company profiles, and user reviews—were embedded in a single application. While this structure offered simplicity, it became a bottleneck as the platform grew. To improve scalability and maintainability, the system was redesigned using independent services for job, company, and review management. Each service now operates with its own database and deployment pipeline, enabling fault isolation and independent scaling.

The schema of the project is designed to represent the core entities: Job, Company, and Review. In the monolithic version, these entities share a centralised database. However, in the microservices architecture, each service manages its own database, ensuring data autonomy and service independence.

The transition

The transition from a monolithic to a microservices architecture involved carefully decoupling the tightly integrated components of the original system. Initially, all business logic for job postings, company information, and user reviews resided within a single codebase and shared a common database, as illustrated in Figure 1. To enable independent development and deployment, each core feature was extracted into its own service as depicted in Figure 2, with separate repositories and databases. APIs were introduced for inter-service communication, replacing direct

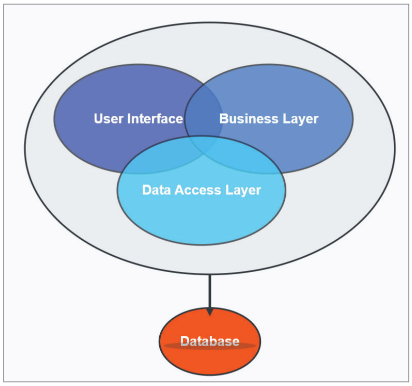

Figure 1: Monolithic architecture